Taewoong Kim

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Research Interests

Embodied AI, World action models, LLM-as-policy, Multimodal learning, Few-shot/zero-shot learning

Education

Yonsei University *Mar. 2023 – Aug. 2024*
M.S. in Artificial Intelligence (Advisor: Prof. Jonghyun Choi)

- Overall GPA: 4.04/4.3
- Relevant Coursework: Multimodal Deep Learning, Machine Learning and Programming

Yonsei University *Mar. 2016 – Aug. 2022*
B.S. in Mechanical Engineering

- Major GPA: 4.16/4.3, Overall GPA: 3.83/4.3
- Relevant Coursework: Dynamics, Mechatronics, Mechanical System Control, Intelligent Control

Work Experience

Samsung Electronics **Seoul, Korea**
Researcher, Samsung Research *Aug. 2025 – Present*

- Research on world action models (WAMs) and LLM-as-policy approaches for robot control

Samsung Electronics **Suwon, Korea**
Engineer, Robot Business Team *Sep. 2022 – Jan. 2023*

- Designed and developed mechanical components for an exoskeleton robot

Samsung Electronics **Suwon, Korea**
Intern, Mobile Experience Division *Jul. 2021 – Aug. 2021*

- Conducted a project on foldable mobile devices

Research Papers

Multi-Modal Grounded Planning and Efficient Replanning For Learning Embodied Agents with A Few Examples AAAI 2025 (Oral)
T. Kim, *B. Kim*, *J. Choi* Project Paper

ReALFRED: An Embodied Instruction Following Benchmark in Photo-Realistic Environments ECCV 2024
T. Kim*, *C. Min**, *B. Kim*, *J. Kim*, *W. Jeong*, *J. Choi* Project Paper

ECLAIR: Event-Cognizant Language Interaction Embodied Robots RO-MAN 2023 Workshop
J. Kim, *B. Kim*, *C. Min*, *Y. Kim*, ***T. Kim***, *J. Choi*

** Equal contribution*

Patents

Method for Performing Tasks According to FLARE Model Including Multi-Modal Planning Module and Environment Adaptive Replanning Module and AI Agent Using the Same KR 10-2024-0166732
T. Kim, *B. Kim*, *J. Choi* *Filed Nov. 2024 (Pending)*

Research Experience

SNU Machine Perception and Reasoning Lab **Seoul National University**
Research Assistant (Advisor: Prof. Jonghyun Choi) *Sep. 2024 – Mar. 2025*

- **Zero-shot affordance grounding**
  - Leveraged image generation models for zero-shot affordance grounding

- Developed a framework that treats occlusions as interaction signals to identify affordance regions

Yonsei Vision & Learning Lab **Yonsei University**
Graduate Research Assistant (Advisor: Prof. Jonghyun Choi) *Mar. 2023 – Aug. 2024*

- **Large language models (LLMs) as few-shot planners for embodied agents**
  - Developed a multimodal LLM-based planner for enhanced grounded planning
  - Designed an efficient replanning system that corrects partially misleading subgoals
- **Bridging the reality gap**
  - Proposed a photo-realistic benchmark built on 3D-captured indoor scenes with interactive objects

Honors and Awards

Outstanding Paper Award (Golden Prize), IPIU 2025 *Feb. 2025*
Innovative Graduate Thesis Award, Yonsei University *Feb. 2025*
Outstanding Paper Award (Silver Prize), IPIU 2024 *Feb. 2024*
1st Place Award, CVPRW 2023 Embodied AI Workshop Challenge *June 2023*
Academic High Honors Award, Yonsei University *Fall 2019, Fall 2020*
Academic Honors Award, Yonsei University *Spring 2019*
Academic Scholarship, Gwacheon City *Spring 2020 – Fall 2020*
Volunteer Scholarship, Yonsei University *Spring 2019 – Fall 2019*
Veritas (Academic) Scholarship, Yonsei University *Spring 2019, Fall 2019, Spring 2020, 2021, 2022*

Leadership and Service

Korean University Mentors Union *Sep. 2020 – Feb. 2021*
Student Mentor

School of Mechanical Engineering, Yonsei University *Dec. 2018 – Dec. 2019*
56th Student President
- Led the department student council and represented students in department decisions

Republic of Korea Army *Apr. 2017 – Jan. 2019*
Sergeant, Mandatory Military Service

Teaching Experience

Teaching Assistant, Seoul National University *Fall 2024*
TA for graduate AI seminar

Academic Services

Reviewer: AAAI'27, BMVC'26, ICCV'25, RO-MAN'24

Skills

Languages	Korean (Native), English (Fluent)
Programming	Python, C/C++, MATLAB
CAD/CAE	NX, Ansys, PTC Creo
Tools	Git/GitHub, LaTeX